

PHYSICS

Vectors in Physics Laws & Multiplying Vectors

Dot Product · Cross Product · Laws of Motion · Applications

1. Vectors in the Laws of Physics

Many fundamental laws of physics are expressed using vectors. Understanding the vector nature of physical quantities is essential for solving real-world problems correctly.

1.1 Newton's First Law — Law of Inertia

Statement: An object remains at rest or moves with constant velocity unless acted upon by a net external force.

Vector Form: $\Sigma \vec{F} = 0 \Rightarrow \vec{v} = \text{constant}$

- $\Sigma \vec{F} = 0$ means the vector sum of all forces is zero — not just magnitudes
- Velocity $\vec{v}$ is a vector: both speed AND direction remain constant
- Example: A book on a table — gravity vector down, normal vector up $\rightarrow$ net = 0

1.2 Newton's Second Law — $\vec{F} = m\vec{a}$

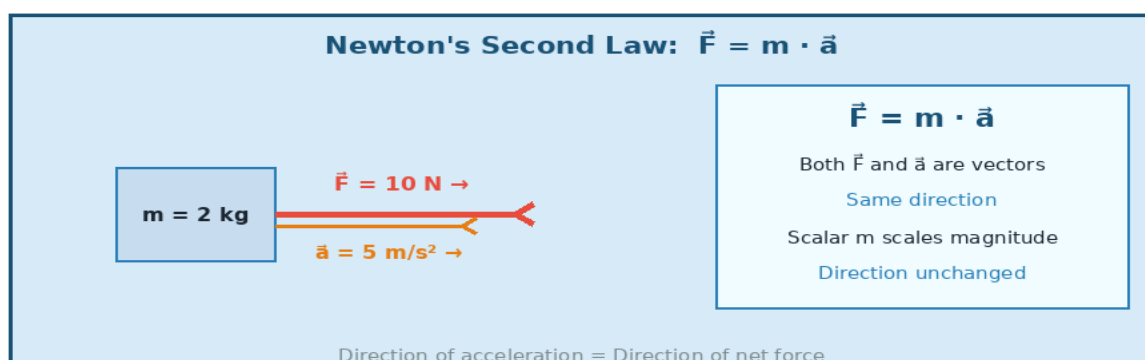


Figure 1: Newton's 2nd Law — net force $\vec{F}$ and acceleration $\vec{a}$ are both vectors in the same direction

Statement: The net force on an object equals its mass times its acceleration.

$$\vec{F} = m \cdot \vec{a}$$

m is scalar; $\vec{F}$ and $\vec{a}$ are vectors

- Direction of acceleration = Direction of net force (always)
- Mass m is a positive scalar — it scales magnitude but does NOT change direction
- If $\vec{F}$ points East, $\vec{a}$ also points East
- Components: $F_x = m a_x$, $F_y = m a_y$, $F_z = m a_z$

1.3 Work — $W = \vec{F} \cdot \vec{d}$ (Dot Product)

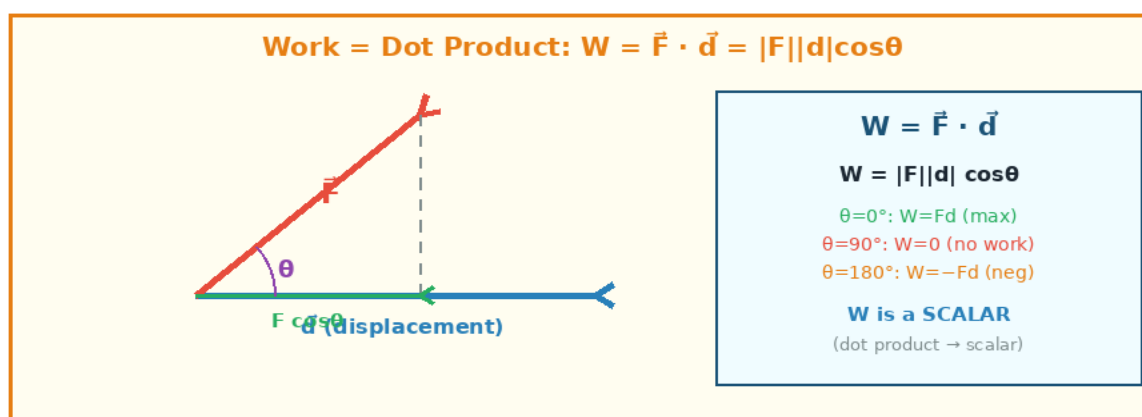


Figure 2: Work = dot product of force and displacement — only the component of F along $\vec{d}$ does work

Work is a scalar quantity obtained from the dot product of force and displacement vectors:

$$W = \vec{F} \cdot \vec{d} = |\vec{F}||\vec{d}|\cos\theta$$

θ = angle between $\vec{F}$ and $\vec{d}$

$\theta = 0^\circ$: Force parallel to motion $\rightarrow W = Fd$ (Maximum positive work)

$\theta = 90^\circ$: Force perpendicular to motion $\rightarrow W = 0$ (No work done)

$\theta = 180^\circ$: Force opposing motion $\rightarrow W = -Fd$ (Negative work)

- Work is a SCALAR — the result of a dot product is always a scalar

1.4 Torque — $\vec{\tau} = \vec{r} \times \vec{F}$ (Cross Product)

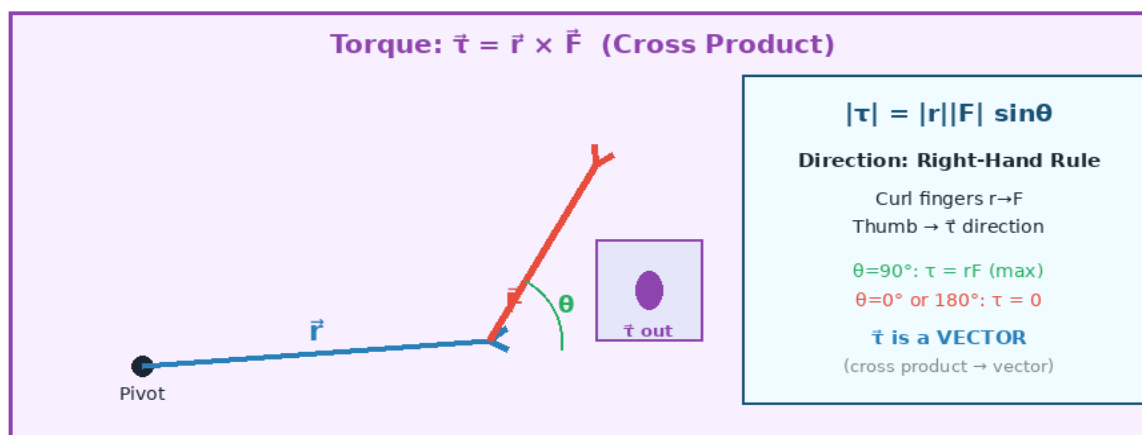


Figure 3: Torque is the cross product of position $\vec{r}$ and force $\vec{F}$ — direction found by right-hand rule

Torque is the rotational effect of a force. It is a vector obtained by the cross product:

$$\vec{\tau} = \vec{r} \times \vec{F}, \quad |\tau| = |\mathbf{r}||\mathbf{F}| \sin\theta$$

$\vec{r}$ = position vector from pivot to point of force application

- $\vec{\tau}$ is perpendicular to both $\vec{r}$ and $\vec{F}$ (given by right-hand rule)
- Maximum torque when $\theta = 90^\circ$ (force perpendicular to $\vec{r}$)
- Zero torque when force is parallel/anti-parallel to $\vec{r}$ ($\theta = 0^\circ$ or 180°)
- Units: $\text{N}\cdot\text{m}$ | Direction: into or out of the plane

1.5 Magnetic Force — $\vec{F} = q(\vec{v} \times \vec{B})$

The force on a moving charge in a magnetic field is a cross product:

$$\vec{F} = q(\vec{v} \times \vec{B})$$

q = charge, $\vec{v}$ = velocity, $\vec{B}$ = magnetic field

- $\vec{F}$ is always perpendicular to both $\vec{v}$ and $\vec{B}$
- No work done by magnetic force ($\vec{F} \perp \vec{v}$, so $W = \vec{F} \cdot \vec{v} = 0$)
- Positive charge: use right-hand rule; Negative charge: reverse direction
- Example: Electron in a field $\rightarrow$ circular or helical motion

1.6 Angular Momentum — $\vec{L} = \vec{r} \times \vec{p}$

Angular momentum is a cross product of position and linear momentum:

$$\mathbf{L}^{\rightarrow} = \mathbf{r}^{\rightarrow} \times \mathbf{p}^{\rightarrow} = \mathbf{r}^{\rightarrow} \times (m\mathbf{v}^{\rightarrow})$$

$\mathbf{p}^{\rightarrow} = m\mathbf{v}^{\rightarrow}$ (linear momentum)

- $\mathbf{L}^{\rightarrow}$ is conserved when net torque is zero (Law of Conservation of Angular Momentum)
- $|\mathbf{L}| = |\mathbf{r}||\mathbf{p}| \sin\theta = mrv \sin\theta$
- Direction: perpendicular to the plane of $\mathbf{r}^{\rightarrow}$ and $\mathbf{p}^{\rightarrow}$

1.7 Velocity in Circular Motion — $\mathbf{v}^{\rightarrow} = \boldsymbol{\omega}^{\rightarrow} \times \mathbf{r}^{\rightarrow}$

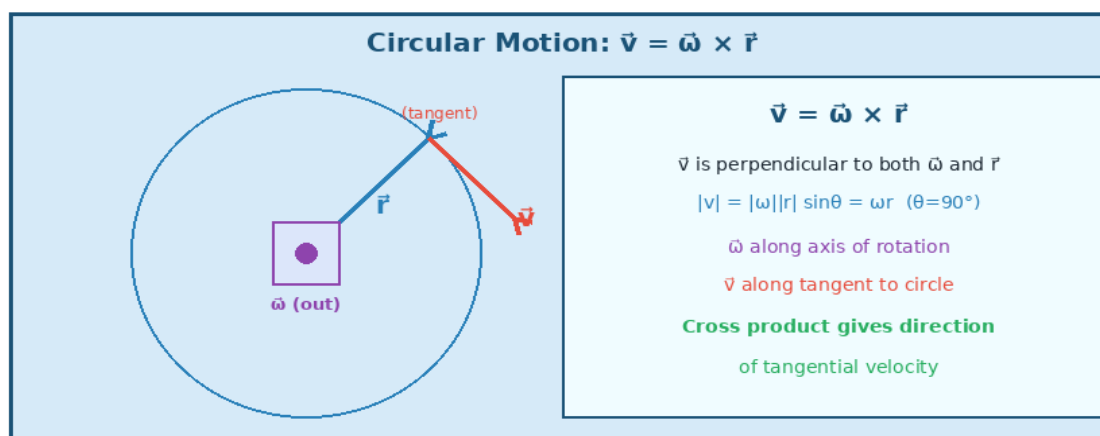


Figure 4: Circular motion — velocity $\mathbf{v}^{\rightarrow} = \boldsymbol{\omega}^{\rightarrow} \times \mathbf{r}^{\rightarrow}$ is always tangent to the circle

$$\mathbf{v}^{\rightarrow} = \boldsymbol{\omega}^{\rightarrow} \times \mathbf{r}^{\rightarrow}$$

$\boldsymbol{\omega}^{\rightarrow}$ = angular velocity (along rotation axis)

- $\mathbf{v}^{\rightarrow}$ is tangential — always perpendicular to $\mathbf{r}^{\rightarrow}$
- $|\mathbf{v}| = \omega r$ (since $\theta = 90^\circ$ between $\boldsymbol{\omega}^{\rightarrow}$ and $\mathbf{r}^{\rightarrow}$)
- $\boldsymbol{\omega}^{\rightarrow}$ points along the axis of rotation (right-hand rule)

2. Scalar Multiplication of a Vector

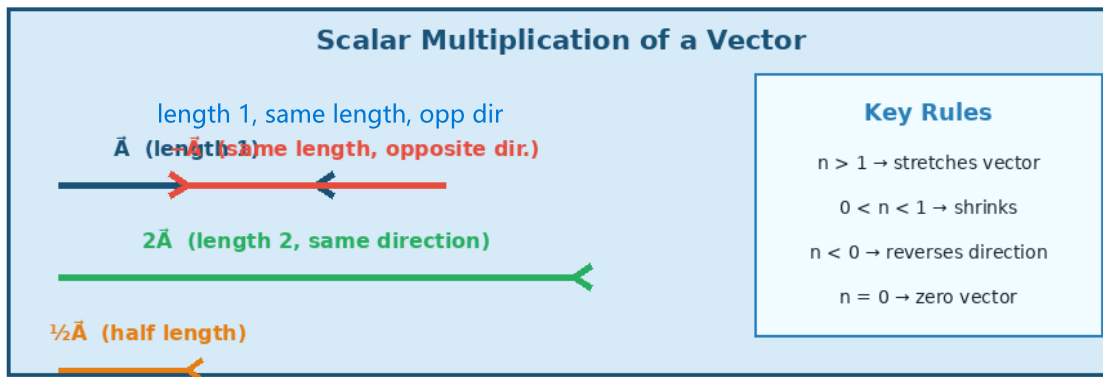


Figure 5: Scalar multiplication — stretches, shrinks, or reverses a vector

When a vector is multiplied by a scalar (a real number), the magnitude changes but the direction stays the same (or reverses for negative scalars).

$$n \cdot \vec{A} = (nA_x)\hat{i} + (nA_y)\hat{j} + (nA_z)\hat{k}$$

$n = \text{scalar}, \vec{A} = \text{any vector}$

Scalar n	Effect on Vector	Example
$n > 1$	Stretches magnitude	$2 \cdot (3\hat{i}) = 6\hat{i}$
$0 < n < 1$	Shrinks magnitude	$\frac{1}{2} \cdot (6\hat{i}) = 3\hat{i}$
$n < 0$	Reverses direction	$-1 \cdot (5\hat{i}) = -5\hat{i}$
$n = 0$	Zero vector	$0 \cdot \vec{A} = \vec{0}$

3. Dot Product (Scalar Product)

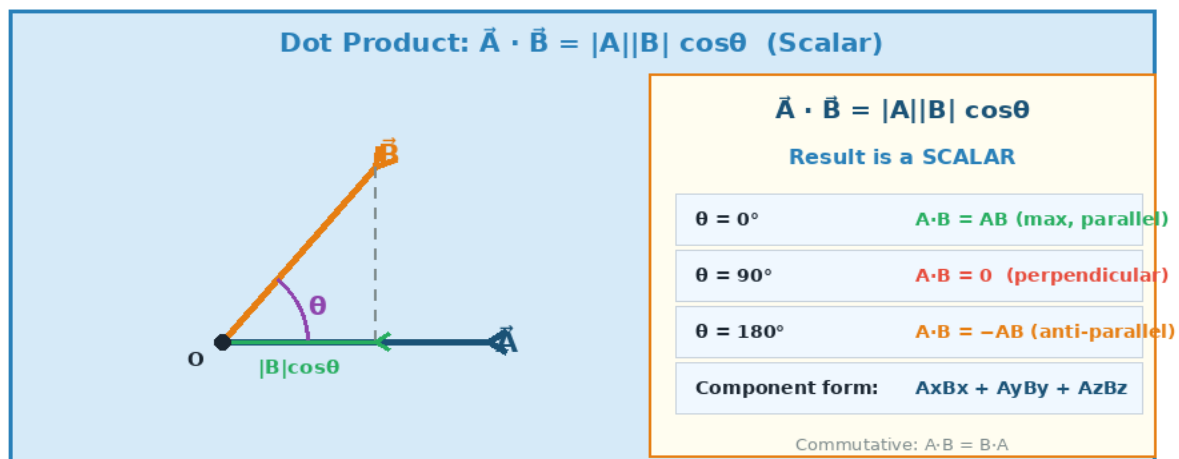


Figure 6: Dot product — $\vec{A} \cdot \vec{B}$ equals the magnitude of A times the projection of B onto A

3.1 Definition and Formula

The dot product of two vectors is a SCALAR quantity:

$$\vec{A} \cdot \vec{B} = |\vec{A}| |\vec{B}| \cos\theta$$

θ = angle between the two vectors

$$\vec{A} \cdot \vec{B} = A_x B_x + A_y B_y + A_z B_z$$

Component form (most useful for calculation)

3.2 Properties of Dot Product

Property	Formula	Meaning
Commutative	$\vec{A} \cdot \vec{B} = \vec{B} \cdot \vec{A}$	Order doesn't matter
Distributive	$\vec{A} \cdot (\vec{B} + \vec{C}) = \vec{A} \cdot \vec{B} + \vec{A} \cdot \vec{C}$	Distributes over addition
Self dot	$\vec{A} \cdot \vec{A} = \vec{A} ^2$	Magnitude squared
Perpendicular	$\vec{A} \cdot \vec{B} = 0$ if $\theta = 90^\circ$	Orthogonality test
Parallel	$\vec{A} \cdot \vec{B} = \pm \vec{A} \vec{B} $ if $\theta = 0^\circ / 180^\circ$	Max/min value
Unit vectors	$\hat{i} \cdot \hat{i} = \hat{j} \cdot \hat{j} = \hat{k} \cdot \hat{k} = 1$	Self dot = 1
Diff. unit vec.	$\hat{i} \cdot \hat{j} = \hat{j} \cdot \hat{k} = \hat{k} \cdot \hat{i} = 0$	Perpendicular

3.3 Physical Applications of Dot Product

Application	Formula	Description
Work	$W = \vec{F} \cdot \vec{d}$	Energy transferred by a force
Power	$P = \vec{F} \cdot \vec{v}$	Rate of work done
Electric Flux	$\Phi = \vec{E} \cdot \vec{A}$	Field lines through a surface
Projection	$\text{proj} = \vec{A} \cdot \hat{B}$	Shadow of A onto B direction
Angle between	$\cos\theta = \vec{A} \cdot \vec{B} / (\vec{A} \vec{B})$	Find angle between vectors

4. Cross Product (Vector Product)

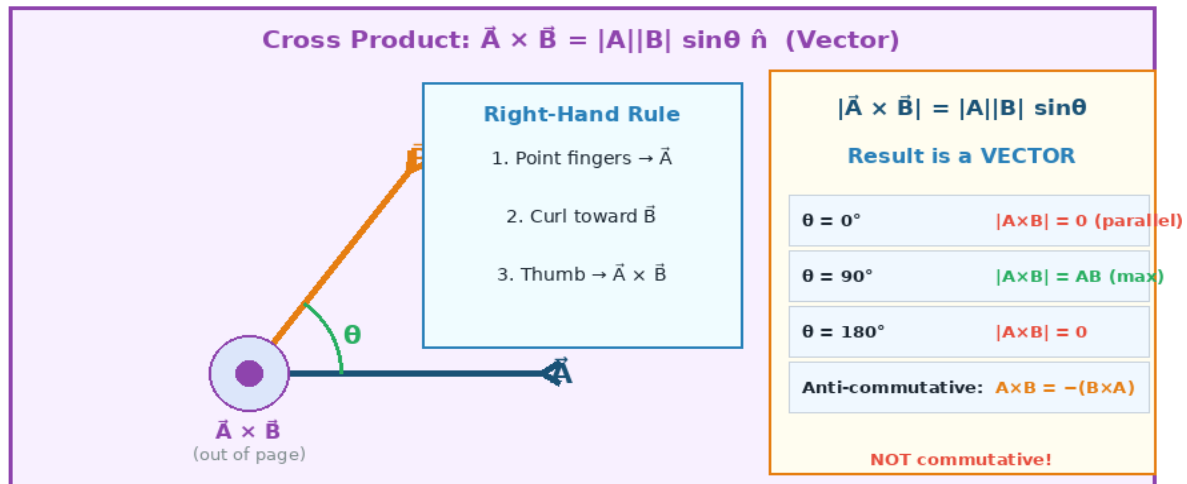


Figure 7: Cross product — $\vec{A} \times \vec{B}$ gives a vector perpendicular to both, with right-hand rule for direction

4.1 Definition and Formula

The cross product of two vectors produces a VECTOR perpendicular to both:

$$\vec{A} \times \vec{B} = |\vec{A}| |\vec{B}| \sin\theta \cdot \hat{n}$$

$\hat{n}$ = unit vector perpendicular to both (right-hand rule)

Determinant (Component) Form:

$$\vec{A} \times \vec{B} = (A_y B_z - A_z B_y) \hat{i} - (A_x B_z - A_z B_x) \hat{j} + (A_x B_y - A_y B_x) \hat{k}$$

4.2 Properties of Cross Product

Property	Formula	Meaning
Anti-commutative	$\vec{A} \times \vec{B} = -(\vec{B} \times \vec{A})$	Order matters!
Distributive	$\vec{A} \times (\vec{B} + \vec{C}) = \vec{A} \times \vec{B} + \vec{A} \times \vec{C}$	Distributes
Self cross	$\vec{A} \times \vec{A} = \vec{0}$	Parallel vectors $\rightarrow$ zero
Perpendicular	$\vec{A} \times \vec{B} \perp \vec{A}$ and $\perp \vec{B}$	Always true
Unit cross	$\hat{i} \times \hat{j} = \hat{k}$, $\hat{j} \times \hat{k} = \hat{i}$	Cyclic pattern
	$\hat{k} \times \hat{i} = \hat{j}$, $\hat{j} \times \hat{i} = -\hat{k}$	Reverse = negative

4.3 Right-Hand Rule

To find the direction of $\vec{A} \times \vec{B}$:

- Step 1: Point your fingers in the direction of $\vec{A}$
- Step 2: Curl them toward $\vec{B}$ (through the smaller angle θ)
- Step 3: Your thumb points in the direction of $\vec{A} \times \vec{B}$
- Note: $\vec{B} \times \vec{A}$ is in the OPPOSITE direction (use left hand or negate)

4.4 Physical Applications of Cross Product

Application	Formula	Description
Torque	$\vec{\tau} = \vec{r} \times \vec{F}$	Rotational effect of force
Magnetic Force	$\vec{F} = q(\vec{v} \times \vec{B})$	Force on moving charge
Angular Momentum	$\vec{L} = \vec{r} \times \vec{p}$	Rotational momentum
Velocity (circular)	$\vec{v} = \vec{\omega} \times \vec{r}$	Tangential velocity
Area of parallelogram	$ \vec{A} \times \vec{B} $	Geometric area

5. Dot Product vs Cross Product — Comparison

Dot Product vs Cross Product — Quick Comparison	
DOT PRODUCT $\vec{A} \cdot \vec{B}$	CROSS PRODUCT $\vec{A} \times \vec{B}$
Result type: Scalar	Result type: Vector
Formula: $ \vec{A} \vec{B} \cos\theta$	Formula: $ \vec{A} \vec{B} \sin\theta \hat{n}$
$\theta = 0^\circ$: Maximum (AB)	$\theta = 0^\circ$: Zero (0)
$\theta = 90^\circ$: Zero (0)	$\theta = 90^\circ$: Maximum (AB)
Commutative?: Yes: $\vec{A} \cdot \vec{B} = \vec{B} \cdot \vec{A}$	Commutative?: No: $\vec{A} \times \vec{B} = -\vec{B} \times \vec{A}$
Physics use: Work, Power, Flux	Physics use: Torque, Force ($\vec{B}$), $\vec{\omega} \times \vec{r}$

Figure 8: Side-by-side comparison of dot and cross product properties

DOT PRODUCT ($\cdot$)	CROSS PRODUCT ($\times$)
$\vec{A} \cdot \vec{B} = \vec{A} \vec{B} \cos\theta$ - Result is a SCALAR - Maximum at $\theta = 0^\circ$ (parallel) - Zero at $\theta = 90^\circ$ (perpendicular) - Commutative: $\vec{A} \cdot \vec{B} = \vec{B} \cdot \vec{A}$ - Component form: $A_x B_x + A_y B_y + A_z B_z$	$\vec{A} \times \vec{B} = \vec{A} \vec{B} \sin\theta \hat{n}$ - Result is a VECTOR - Maximum at $\theta = 90^\circ$ (perpendicular) - Zero at $\theta = 0^\circ$ (parallel) - Anti-commutative: $\vec{A} \times \vec{B} = -\vec{B} \times \vec{A}$ - Uses determinant for components

- Uses: Work, Power, Flux, Angle

- Uses: Torque, Force(B), $L^{\vec{}}$, $v=\omega \times r$

6. Sample Problems with Solutions

Sample Problem 1 — Work Done by a Force (Dot Product)

Question: A force $\vec{F} = 50 \text{ N}$ acts at 37° above the horizontal. An object moves 10 m horizontally. Find the work done.

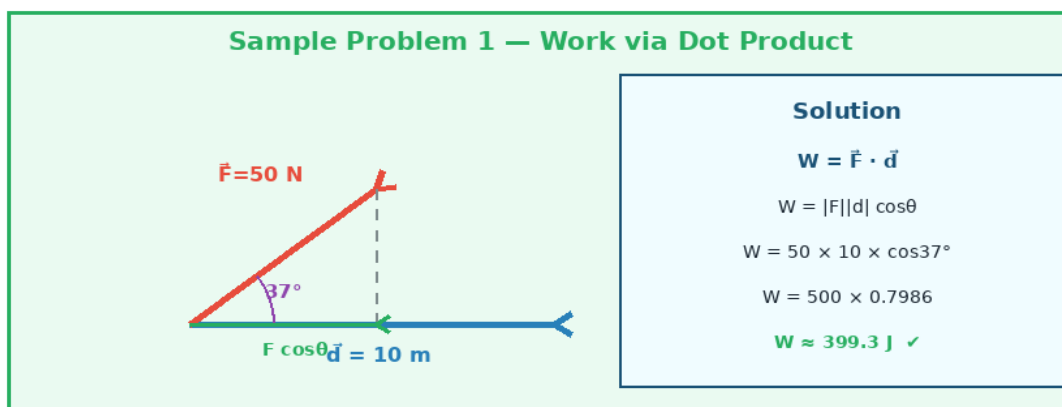


Figure 9: Sample Problem 1 — only the horizontal component of F does work

✓ SOLUTION

Given: $|\vec{F}| = 50 \text{ N}$, $|\vec{d}| = 10 \text{ m}$, $\theta = 37^\circ$

Formula	$W = \vec{F} \cdot \vec{d} = \vec{F} \vec{d} \cos \theta$
Substitute	$W = 50 \times 10 \times \cos 37^\circ$
cos 37°	$\cos 37^\circ = 0.7986$
Calculate	$W = 500 \times 0.7986 = 399.3 \text{ J}$
Answer	$W \approx 399.3 \text{ J} \checkmark$

Note: Only the component of force in the direction of displacement ($F \cos \theta = 50 \times 0.7986 = 39.93 \text{ N}$) does work. The perpendicular component ($F \sin \theta = 50 \times 0.6018 = 30.09 \text{ N}$) does zero work.

Sample Problem 2 — Torque (Cross Product)

Question: A wrench applies a force $\vec{F} = 30\text{ N}$ downward at the end of a handle $\vec{r} = 0.4\text{ m}$ horizontal. Find the torque magnitude and direction.

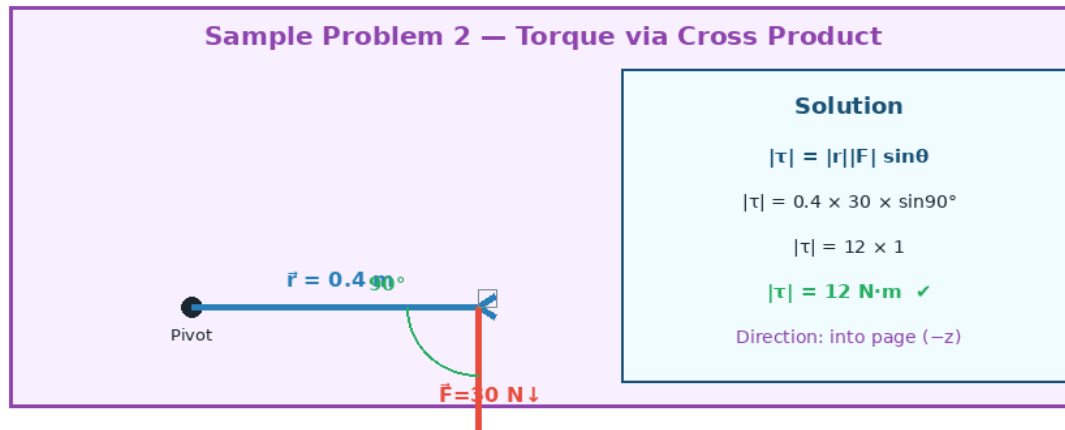


Figure 10: Sample Problem 2 — torque $= \vec{r} \times \vec{F}$ with 90° between them

✓ SOLUTION

Given: $|\vec{r}| = 0.4\text{ m}$ (East), $|\vec{F}| = 30\text{ N}$ (downward), $\theta = 90^\circ$

Formula	$ \tau = \vec{r} \vec{F} \sin\theta$
Substitute	$ \tau = 0.4 \times 30 \times \sin 90^\circ$
$\sin 90^\circ$	$\sin 90^\circ = 1.000$
Calculate	$ \tau = 12 \times 1 = 12\text{ N}\cdot\text{m}$
Magnitude	$ \tau = 12\text{ N}\cdot\text{m} \quad \checkmark$
Direction	$\vec{\tau} = \vec{r} \times \vec{F}$: East $\times$ Down = Into page ($-z$) $\checkmark$

Sample Problem 3 — Dot Product Using Components

Question: Given $\vec{A} = 3\hat{i} + 4\hat{j}$ and $\vec{B} = 2\hat{i} - \hat{j} + 2\hat{k}$, find (a) $\vec{A} \cdot \vec{B}$, (b) angle between them.

✓ SOLUTION

Given: $\vec{A} = 3\hat{i} + 4\hat{j} + 0\hat{k}$, $\vec{B} = 2\hat{i} - \hat{j} + 2\hat{k}$

(a) Formula	$\vec{A} \cdot \vec{B} = A_x B_x + A_y B_y + A_z B_z$
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Substitute	$\vec{A} \cdot \vec{B} = (3)(2) + (4)(-1) + (0)(2)$
Calculate	$\vec{A} \cdot \vec{B} = 6 - 4 + 0$
Result	$\vec{A} \cdot \vec{B} = 2 \quad \checkmark$
(b) A 	$ A = \sqrt{3^2 + 4^2 + 0^2} = \sqrt{25} = 5$
 B 	$ B = \sqrt{2^2 + 1^2 + 2^2} = \sqrt{9} = 3$
cosθ	$\cos\theta = (\vec{A} \cdot \vec{B}) / (A B) = 2 / (5 \times 3) = 2/15$
θ	$\theta = \cos^{-1}(2/15) = \cos^{-1}(0.1333) \approx 82.3^\circ \quad \checkmark$

Sample Problem 4 — Cross Product Using Components

Question: Find $\vec{A} \times \vec{B}$ where $\vec{A} = 2\hat{i} + 3\hat{j} + 0\hat{k}$ and $\vec{B} = 1\hat{i} + 0\hat{j} + 4\hat{k}$.

✓ SOLUTION

Given: $\vec{A} = 2\hat{i} + 3\hat{j} + 0\hat{k}$, $\vec{B} = 1\hat{i} + 0\hat{j} + 4\hat{k}$

Using the determinant formula:

$$\vec{A} \times \vec{B} = (A_y B_z - A_z B_y)\hat{i} - (A_x B_z - A_z B_x)\hat{j} + (A_x B_y - A_y B_x)\hat{k}$$

$\hat{i}$ component	$A_y B_z - A_z B_y = (3)(4) - (0)(0) = 12 - 0 = 12$
$\hat{j}$ component	$-(A_x B_z - A_z B_x) = -[(2)(4) - (0)(1)] = -[8 - 0] = -8$
$\hat{k}$ component	$A_x B_y - A_y B_x = (2)(0) - (3)(1) = 0 - 3 = -3$
Result	$\vec{A} \times \vec{B} = 12\hat{i} - 8\hat{j} - 3\hat{k} \quad \checkmark$
Verify $\perp$	$\vec{A} \cdot (\vec{A} \times \vec{B}) = 2(12) + 3(-8) + 0(-3) = 24 - 24 + 0 = 0 \quad \checkmark$

7. Practice Problems

Instructions: Solve each problem fully. Show all steps. Answers are provided at the end.

Practice Problem 1 — Newton's Second Law (Vector)

Three forces act on a 4 kg block:

- $F_1 = 10 \text{ N}$ at 0° (East)
- $F_2 = 8 \text{ N}$ at 90° (North)
- $F_3 = 6 \text{ N}$ at 180° (West)

Find: (a) Net force vector F_{net} , (b) Magnitude $|F_{\text{net}}|$, (c) Direction of F_{net} , (d) Acceleration a

✓ SOLUTION

x-components	$F_{1x}=+10, F_{2x}=0, F_{3x}=-6 \rightarrow \Sigma F_x = 10+0-6 = +4 \text{ N}$
y-components	$F_{1y}=0, F_{2y}=+8, F_{3y}=0 \rightarrow \Sigma F_y = 0+8+0 = +8 \text{ N}$
(a) F_{net}	$F_{\text{net}} = 4\hat{i} + 8\hat{j} \text{ N} \quad \checkmark$
(b) F_{net}	$\sqrt{(4^2+8^2)} = \sqrt{(16+64)} = \sqrt{80} \approx 8.94 \text{ N} \quad \checkmark$
(c) Direction	$\theta = \tan^{-1}(8/4) = \tan^{-1}(2) \approx 63.4^\circ \text{ North of East} \quad \checkmark$
(d) a	$a = F/m = (4\hat{i}+8\hat{j})/4 = 1\hat{i} + 2\hat{j} \text{ m/s}^2 \quad \checkmark$
 a 	$\sqrt{(1^2+2^2)} = \sqrt{5} \approx 2.24 \text{ m/s}^2 \text{ at } 63.4^\circ \text{ N of E} \quad \checkmark$

Practice Problem 2 — Dot Product: Work & Angle

Given two vectors: $P = 4\hat{i} + 3\hat{j} - 2\hat{k}$ and $Q = -1\hat{i} + 5\hat{j} + 3\hat{k}$

Find:

- (a) $P \cdot Q$
- (b) $|P|$ and $|Q|$
- (c) Angle θ between P and Q
- (d) Work done if P is force (N) and Q is displacement (m)

✓ SOLUTION

(a) $P \cdot Q$	$P_x Q_x + P_y Q_y + P_z Q_z = (4)(-1) + (3)(5) + (-2)(3)$ $= -4 + 15 - 6 = +5 \quad \checkmark$
(b) P	$ P = \sqrt{(16+9+4)} = \sqrt{29} \approx 5.385$
Q	$ Q = \sqrt{(1+25+9)} = \sqrt{35} \approx 5.916$
(c) $\cos\theta$	$\cos\theta = 5 / (5.385 \times 5.916) = 5 / 31.86 = 0.157$
θ	$\theta = \cos^{-1}(0.157) \approx 81.0^\circ \quad \checkmark$
(d) Work	$W = P \cdot Q = 5 \text{ J} \quad \checkmark \quad (\text{same as part a})$

Practice Problem 3 — Cross Product & Torque

A force $\vec{F} = 3\hat{i} - 2\hat{j} + 1\hat{k}$ N is applied at position $\vec{r} = 1\hat{i} + 2\hat{j} + 0\hat{k}$ m from a pivot.

Find:

- (a) Torque vector $\vec{\tau} = \vec{r} \times \vec{F}$
- (b) Magnitude $|\vec{\tau}|$
- (c) Verify $\vec{\tau}$ is perpendicular to both $\vec{r}$ and $\vec{F}$

✓ SOLUTION

Given: $\vec{r} = 1\hat{i} + 2\hat{j} + 0\hat{k}$, $\vec{F} = 3\hat{i} - 2\hat{j} + 1\hat{k}$

(a) $\hat{i}$ component	$r_y F_z - r_z F_y = (2)(1) - (0)(-2) = 2 - 0 = 2$
$\hat{j}$ component	$-(r_x F_z - r_z F_x) = -[(1)(1) - (0)(3)] = -[1 - 0] = -1$
$\hat{k}$ component	$r_x F_y - r_y F_x = (1)(-2) - (2)(3) = -2 - 6 = -8$
$\vec{\tau}$	$\vec{\tau} = \vec{r} \times \vec{F} = 2\hat{i} - 1\hat{j} - 8\hat{k} \text{ N}\cdot\text{m} \quad \checkmark$
(b) $\vec{\tau}$	$ \tau = \sqrt{(2^2 + 1^2 + 8^2)} = \sqrt{(4 + 1 + 64)} = \sqrt{69} \approx 8.31 \text{ N}\cdot\text{m} \quad \checkmark$
(c) $\vec{r} \cdot \vec{\tau}$	$\vec{r} \cdot \vec{\tau} = (1)(2) + (2)(-1) + (0)(-8) = 2 - 2 + 0 = 0 \quad \checkmark \perp$
$\vec{F} \cdot \vec{\tau}$	$\vec{F} \cdot \vec{\tau} = (3)(2) + (-2)(-1) + (1)(-8) = 6 + 2 - 8 = 0 \quad \checkmark \perp$

Practice Problem 4 — Magnetic Force (Cross Product)

A proton (charge $q = +1.6 \times 10^{-19}$ C) moves with velocity $\vec{v} = 3 \times 10^6 \hat{i}$ m/s in a magnetic field $\vec{B} = 0.5\hat{j}$ T.

Find:

- (a) The cross product $\vec{v} \times \vec{B}$
- (b) The magnetic force $\vec{F} = q(\vec{v} \times \vec{B})$
- (c) Direction of force on the proton

✓ SOLUTION

Given: $\vec{v} = 3 \times 10^6 \hat{i}$ m/s , $\vec{B} = 0.5\hat{j}$ T , $q = +1.6 \times 10^{-19}$ C

(a) $\vec{v} \times \vec{B}$	$\hat{i} \times \hat{j} = \hat{k}$ (standard unit vector cross product)
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	$\vec{v} \times \vec{B} = (3 \times 10^6) (0.5) (\hat{i} \times \hat{j})$
	$= 1.5 \times 10^6 \hat{k} \text{ m/s} \cdot \text{T} = 1.5 \times 10^6 \hat{k} \text{ m}^2 / (\text{A} \cdot \text{s}^2)$
(b) $\vec{F}$	$\vec{F} = q(\vec{v} \times \vec{B}) = (1.6 \times 10^{-19}) (1.5 \times 10^6 \hat{k})$
	$\vec{F} = 2.4 \times 10^{-13} \hat{k} \text{ N} \quad \checkmark$
(c) Direction	$\hat{k} = +z \text{ direction} = \text{out of page} / \text{upward} \quad \checkmark$

8. Summary & Quick Reference

Law / Concept	Vector Formula	Product Type
Newton's 2nd Law	$\vec{F} = m \cdot \vec{a}$	Scalar $\times$ Vector
Work	$W = \vec{F} \cdot \vec{d}$	Dot ($\rightarrow$ Scalar)
Power	$P = \vec{F} \cdot \vec{v}$	Dot ($\rightarrow$ Scalar)
Torque	$\vec{\tau} = \vec{r} \times \vec{F}$	Cross ($\rightarrow$ Vector)
Magnetic Force	$\vec{F} = q(\vec{v} \times \vec{B})$	Cross ($\rightarrow$ Vector)
Angular Momentum	$\vec{L} = \vec{r} \times \vec{p}$	Cross ($\rightarrow$ Vector)
Circular Velocity	$\vec{v} = \vec{\omega} \times \vec{r}$	Cross ($\rightarrow$ Vector)
Angle Between	$\cos\theta = \vec{A} \cdot \vec{B} / (\vec{A} \vec{B})$	Dot ($\rightarrow$ Scalar)

DOT PRODUCT	CROSS PRODUCT
Key Formula: $\vec{A} \cdot \vec{B} = \vec{A} \vec{B} \cos\theta = A_x B_x + A_y B_y + A_z B_z$ - Result: SCALAR - Max at $\theta=0^\circ$, Zero at $\theta=90^\circ$ - Commutative - Uses: W, P, Flux, angles	Key Formula: $ \vec{A} \times \vec{B} = \vec{A} \vec{B} \sin\theta, \text{ dir: RHR}$ - Result: VECTOR - Max at $\theta=90^\circ$, Zero at $\theta=0^\circ$ - Anti-commutative (sign reverses) - Uses: $\vec{\tau}$, $\vec{F}(B)$, $\vec{L}$, $\vec{v} = \vec{\omega} \times \vec{r}$